Real Gases

Reasons for deviation of real gases from Ideal behaviour:-

In 1873, Vander Walls studied this

He modified the two assumptions in Kinetic Theory, which are not exactly true. These assumptions are:

(i) The molecules are point masses and thus molecules have definite masses but no volume.

(ii) There is no intermolecular attraction in the gases.

In fact, in a real gas, the volume of the bodies of the molecules may be quite appreciable even at ordinary Pr.

Example: At NTP, one gm-mol of a gas, i.e., 6.02 x 10^{23} molecules occupy a volume of 22400 cc.

The diameter of a molecule may be taken as of the order 2×10^{-8} cm. So, the space occupied by a molecule is

$$\frac{4}{3}\pi(2\times10^{-8})^3 = 3.3\times10^{-23} \text{ cc}$$

Hence, the space taken up by the molecules

$$= 6.02\times10^{23}\times3.3\times10^{-23} = 20.46 \text{ cc}$$

Thus at N.T.P the space occupied by the molecules themselves is nearly $\sim \frac{1}{1000}$ of the volume of the gas.

Real Gases

If the pr. be increased to say 10 atm, the volume of the gas would be approximately 2240 cc. Then, the fraction of this volume occupied by the molecule = 20.46/2240 = $\frac{1}{100}$, i.e nearly one percent of the volume, which is no means negligible.

✓**Kinetic Theory presumed that no intermolecular force exists**

↓

✓**But, in fact, there does exist a force of attraction between the molecules**

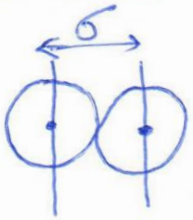
↓

✓**Joule-Thoson's porous-plug experiment definitely proves the existence of intermolecular attraction of the gases**

Definite size of the gas molecules results from the repulsive forces acting on the gas molecules when they are approaching close to make collisions. When two molecules approach to each other, they cannot reach closer together beyond a certain distance σ, called the <u>distance of closest approach</u> and it is also called <u>collision diameter</u>.

Higher the intermolecular repulsion, greater is the value of σ

<u>The Vander Waals radius</u> = $\frac{\sigma}{2}$



Gas molecules have both attraction and repulsion among themselves.

Attractive potential: $\left(V_{attraction} \propto \frac{1}{r^6}\right)$

Repulsive potential $\left(V_{repulsion} \propto \frac{1}{r^{12}}\right)$

Real Gases: Qualitative Explanation of Amagat's Curves

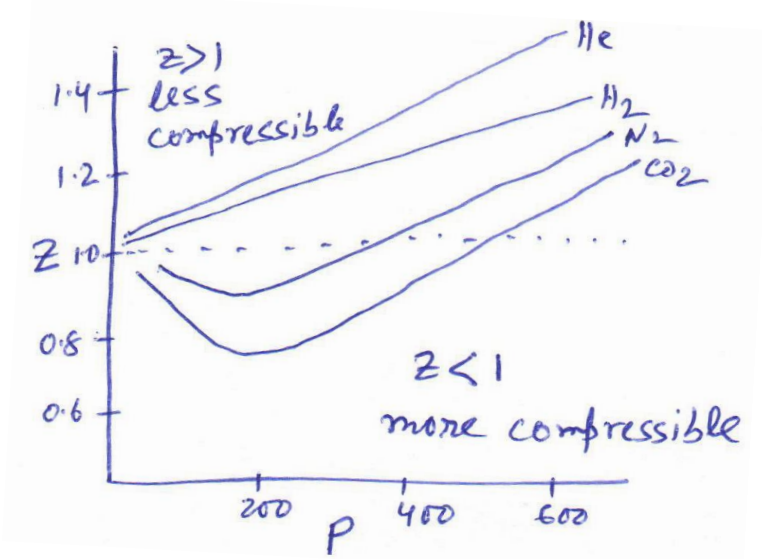
The concept of molecular attraction and repulsion can be used to explain the value of compressibility factor (Z) quantitatively.

(i) At low P ($P \to 0$), the vol. of the gas is large, and intermolecular distance (r) is large, both attractive force and repulsive forces are negligibly small, so it can not affect the ideal behavior. Z becomes one.

$P \to 0$
V is large
r is large

($V_{attraction} \propto \frac{1}{r^6}$)

($V_{repulsion} \propto \frac{1}{r^{12}}$)



(ii) At high P, the r is small, the molecules are very close to each other. At this condition, short-range repulsive potential dominates over the long-range attractive potential. The gas becomes less compressible than ideal gas (where there is no intermolecular interaction) and $Z > 1$.

(iii) At moderate P, the molecules are not very close and long-range attractive potential dominates over the short-range repulsive potential. The gas becomes more compressible and $Z < 1$

For H_2 and He gases, attractive potential is very small in comparison to repulsive potential, thus Z > 1 always except when T is very low (liq. H_2 → -170 ºC and liq. He → -250 ºC).

Real Gases: Equation of States for Real Gases

- ✓ **Many attempts have been made to modify the Ideal Gas Equation**
- ✓ **Changed it into a form which would represent P-V-T relations of Ideal Gases**
- ✓ **These equations are derived on empirical or semi-empirical basis**
- ✓ **The forms of some of these equations are:**

Van der waal's : $\left(P + \frac{a}{V^2}\right)(V-b) = RT$

Dieterici : $P(V-b) = RT\, e^{\frac{-a}{RTV}}$

Berthelot : $\left(P + \frac{a}{TV^2}\right)(V-b) = RT$

⋮

Kammerling onnes : $PV = RT\left(1 + \frac{B}{V} + \frac{C}{V^2} + \cdots\right)$

etc. etc.

Formulation of vander waal's Equation:

In 1873, Vander waal's modified the ideal gas equ. for 1 mole : $P_{id}\bar{V}_{id} = RT$ by incorporating the size effect and intermolecular attraction effect of the real gases. These two effects are discussed under the vol. correction and Pr. correction of the ideal gas Equ..

Real Gases: Formulation of Vander Waal's Equation

Volume Correction:-

In real gas, the molecules suffer strong repulsive force when they come close and collide with each other.

↓

This repulsive force gives rise to definite size of the gas molecules. They have been assumed as rigid spheres.

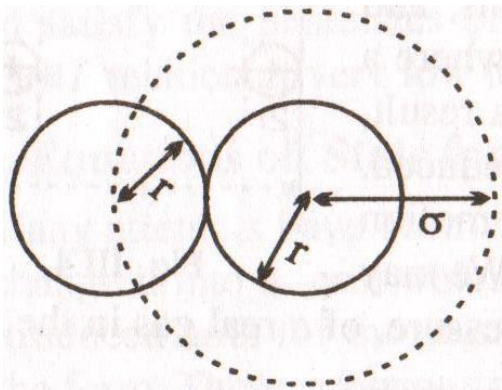
↓

The available vol. for free movement of the molecules in real gas is less than $\bar{V}$.

↓

Let us take available space for free movement of 1 mole gas molecules; $\bar{V}_{id} = \bar{V} - b$

$\bar{V}$ = **Molar vol. of the gas.**
b = Vol. correction term due to definite size of the gas molecules.
$\bar{V}_{id}$ = **Molar vol. of the ideal gas when gas molecules are regarded as point-masses.**



r **= radius of the gas molecule**
$\sigma = 2r$ **= diameter of a molecule of the gas**

- ✓ **When two molecules encounter each other, the distance between the centres of the two molecules = σ.**
- ✓ **These cannot approach closer than this.**
- ✓ **Thus, a space indicated by the dashed circle having a diameter 2σ, will be unavailable to the pair of colliding molecules.**
- ✓ **This space, often called "excluded vol. for a pair of molecules".**

Excluded vol. for a pair of molecules: $\frac{4}{3}\pi\sigma^3$

Real Gases: Formulation of Vander Waal's Equation

Excluded vol. for a pair of molecules: $\frac{4}{3}\pi\sigma^3$

Thus effective vol. of a single molecule $= \frac{1}{2} \times \frac{4}{3}\pi\sigma^3 = \frac{2}{3}\pi\sigma^3$

and $b = \frac{2}{3}\pi N_A \sigma^3$, which is the effective vol. of Avogadro no. of molecules present in 1 mole gas.

Thus, $b = \frac{2}{3}\pi N_A \sigma^3 = 4 \times \frac{4}{3}\pi N_A r^3$ **[σ = 2r]**

So, effective vol., b, is the four times the actual vol. of one mole gas molecules.

The Vander Waal's Eq. after vol. correction:

$$P_{id}(\bar{V} - b) = RT$$

Real Gases: Formulation of Vander Waal's Equation

Pressure Correction:-

Why Pr. of a gas is developed?
Magnitude of Pr. of a gas depends on what?

Pr. exerted by the molecules in the real gas (P) will be less than that if there had not been intermolecular attraction as in the ideal gas (P_{id}).

So, $P_{id} > P \Rightarrow P_{id} = P + P_a$

Here, P_a = Pr. correction term, called internal Pr.
*** It originates from attraction forces between the molecules.**
*** Higher the intermolecular attraction in a gas, greater is the magnitude of P_a**

The reduction of Pr. (P_a) acts with magnitude proportional to the density of the gas molecules.

Both the frequency and force of each collisions are reduced by intermolecular attraction, so the Pr. is reduced in proportional to square of the gas density.

Density of a gas $\propto \frac{1}{\bar{V}}$ **So,** $P_a \propto \frac{1}{\bar{V}^2}$; or $P_a = \frac{a}{\bar{V}^2}$; **a =** Constant for the gas
= Measure of intermolecular attraction of the gas

Incorporating both the vol. correction and Pr. correction, the equn. formed is called Vander Wall's equn. for 1 mole gas

$$\left(P + \frac{a}{\bar{V}^2}\right)(\bar{V} - b) = RT$$

To convert the equn. for n moles, $\bar{V}$ is to be replaced by V/n, where V is the vol. of n moles of the gas. Thus, Vander waals equn. for n mole real gas.

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

Real Gases: Vander Waal's Equation

Vander Waal's Equation (VDW) for *n* mole real gas:

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

Units of "a" and "b" in the VDW Eq.:-

From the VDW Equ.; $P_a = \frac{an^2}{V^2} \Rightarrow a = \frac{P_a \times V^2}{n^2}$

So, unit of $a = \frac{atm \times L^2}{mol^2} = atm\ L^2\ mol^{-2}$

Again, nb = unit of vol.

So, $b = L\ mol^{-1}$

Significance of "*a*" and "*b*" :-

- ✓ **"*a*" term originates from the intermolecular attraction, thus "*a*" is a measure of internal Pr. of the gas**
- ✓ **It measures the attractive forces between the molecules**
- ✓ **Higher the value of "*a*", greater is the intermolecular attraction & more easily the gas could be liquified**

$a_{CO_2} = 3.95\ atm\ L^2\ mol^{-2}$ and $a_{H_2} = 0.22\ atm\ L^2\ mol^{-2}$.

So, CO_2 is more easily liquefied than H_2 gas.

- ✓ **"*b*" measures the molecular size and also a measure of repulsive forces**
- ✓ **The value of "*b*" can be utilized to calculate the molecular diameter, σ**
- ✓ **The greater the value of "*b*", larger is the size of the gas molecule**

Thus, $b_{CO_2} = 0.04\ L\ mol^{-1}$ and $b_{H_2} = 0.02\ L\ mol^{-1}$